

Markov Decision Processes

- States s in S : everything the agent needs to know about the world right now
- Actions a in A : the choices available in each state
- Reward $R(s, a)$: a number received for taking action a in state s
- Transition function $T(s, a, s')$: where you land after each action
- Discount factor γ in $[0, 1)$: how much later rewards count

The Bellman Equation

Equation

$$V(s) = \max_a [R(s, a) + \gamma \sum_{s'} T(s, a, s') V(s')]$$

$V(s)$: value of state s

$\max_a$: best action available

$\sum_{s'} T(s, a, s')$: expectation over next states

γ : discount factor

Intuition

Value = immediate reward + discounted future

Recursive: V appears on both sides

Exactly one fixed point

Solve it by repeated substitution

Everything else in the course builds on this

Value Iteration

1. Initialise $V_0(s) = 0$ for every state
2. Sweep: $V_{k+1}(s) = \max_a [R(s, a) + \gamma \sum_{s'} T(s, a, s') V_k(s')]$
3. Stop when $\max_s |V_{k+1}(s) - V_k(s)| < \epsilon$
4. Read off the greedy policy $\pi(s) = \arg\max_a [\dots]$

